



CEP-CCIT
FAKULTAS TEKNIK

Implementation of Academic Information System Using Python and MySQL

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CEP CCIT

FAKULTAS TEKNIK UNIVERSITAS INDONESIA

2025

PROJECT ON

Implementation of Academic Information System Using Python and MySQL

Developed by

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Implementation of Academic Information System Using Python and MySQL

Batch Code : 3CS1

Start Date : September 11th, 2025

End Date : September 26th, 2025

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Date of Submission: September 26th, 2025

ACKNOWLEDGEMENT

First and foremost, I would like to express my deepest gratitude to Allah SWT for His endless blessings and guidance that have enabled me to complete this project, titled *“Implementation of Academic Information System Using Python and MySQL”*

I would like to extend my sincere appreciation to my supervisor, Mr. Ivan Virdaus S.T & Mr. Tri Agus Riyadi, S.Kom., M.T., for their invaluable support, encouragement, and insightful guidance throughout the development of this project. Their expertise and patience played a crucial role in shaping this work.

My heartfelt thanks go to my family and friends for their unwavering support, prayers, and motivation. Their belief in my abilities has always been a source of strength and inspiration.

I am also grateful to my colleagues and everyone who provided constructive feedback, technical assistance, and moral support during the completion of this project.

Finally, my appreciation goes to all those who have directly or indirectly contributed to the success of this work. Thank you for being part of this journey.

Depok, 29 September 2025

BACKGROUND

As technology continues to evolve, its role in assisting and facilitating human activities becomes increasingly important. One of the most widely recognized technologies is the information system, which continues to develop and gradually influence everyday life. Tasks that were previously done manually and required a lot of energy can now be completed more easily thanks to these developments.

In educational institutions, one of the most noticeable technological developments is in academic data management. Manual management of student, lecturer, course, and Study Plan Card (KRS) data is often inefficient and prone to errors. To overcome this problem, it is necessary to implement a technological system that can manage data in a structured and efficient manner.

This project aims to develop a simple academic information system built to manage student data and grades per semester. This system is designed using the Python programming language and integrated with a MySQL database through XAMPP/phpMyAdmin. With a multi-user concept, this system has two types of users, namely admin and students, with different access rights to demonstrate the multi-user concept application in an academic information system.

This report will focus on how this platform works, using Python and MySQL. In addition, this report will also present flowcharts and entity-relationship diagrams (ERDs) to facilitate understanding of the flow and relationships of data within the system.

SYSTEM ANALYSIS

System Summary :

This project is a simple academic information system built to manage student data and grades per semester. The system was developed using the Python programming language and integrated with a MySQL database through XAMPP/phpMyAdmin. The system is designed with two types of users, namely administrators and students, who have different access rights. This project demonstrates the application of multi-user concepts and integration between Python and databases for academic information system needs.

User Functions :

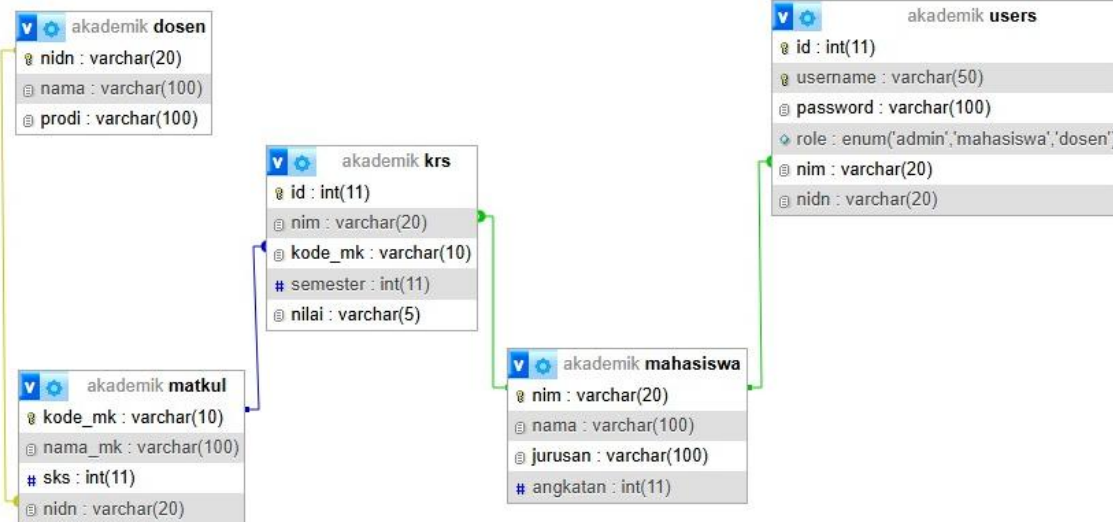
1. Admin :

- Logs in using a pre-defined username and password.
- Manages student data through simple CRUD (Create, Read, Update, Delete) features, such as displaying all student data and adding new students.
- Automatically creates a login account for each new student, with their **NIM** (Student ID number) serving as both the username and password.

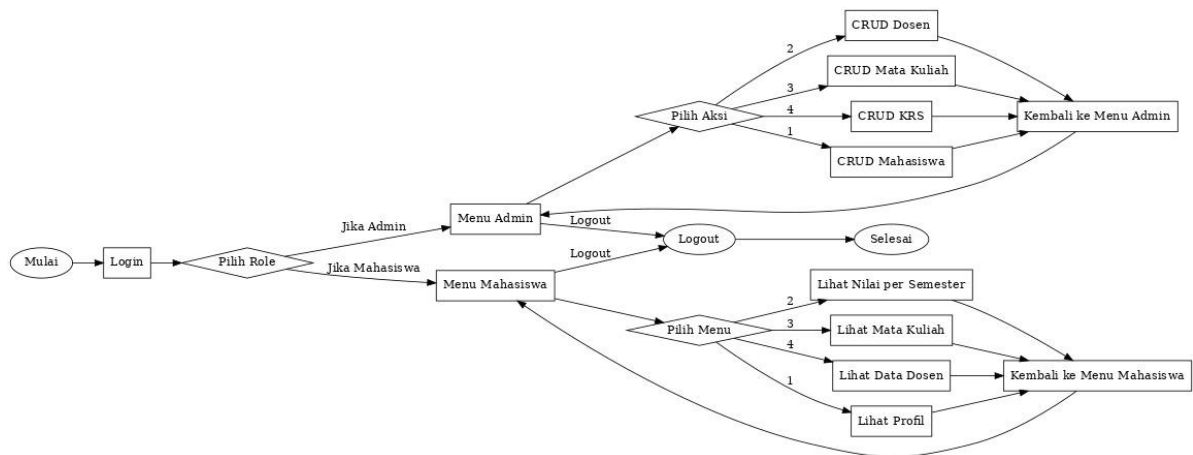
2. Student :

- Logs in using their **NIM**.
- Upon successful login, receives a welcome message : “Welcome, [Student Name] (NIM: [NIM])”.
- Has two main features : viewing their profile and viewing grades per semester, with grades displayed based on KRS (Course Registration Form) data linked to courses and semesters.

ENTITY-RELATIONSHIP DIAGRAM (ERD)

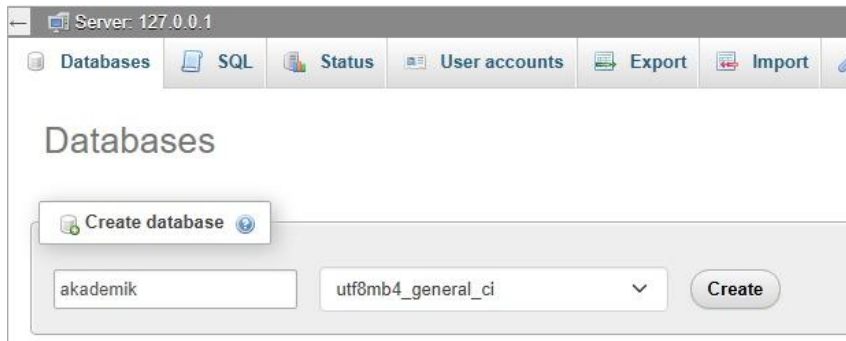


FLOWCHART



MYSQL DATABASE CONFIGURATION

1. The first step is to access phpMyAdmin after the XAMPP MySQL service is running. In the **Create database** field, enter the academic name and select the collation **utf8mb4_general_ci**, then click **Create**. After that, the academic database is ready to use.



2. Next, create each table according to the designed schema:

- Creating a **mahasiswa** table

```
1 CREATE TABLE mahasiswa (  
2   nim VARCHAR(10) PRIMARY KEY,  
3   nama VARCHAR(100) NOT NULL,  
4   jurusan VARCHAR(50) NOT NULL,  
5   angkatan VARCHAR(10) NOT NULL,  
6   no_hp VARCHAR(15)  
7 );
```

- Creating a **dosen** table

```
1 CREATE TABLE dosen (  
2   nidn VARCHAR(10) PRIMARY KEY,  
3   nama_dosen VARCHAR(100) NOT NULL,  
4   prodi VARCHAR(50) NOT NULL  
5 );  
6 |
```

MYSQL DATABASE CONFIGURATION

- Creating a **mata_kuliah** table

```
1 CREATE TABLE mata_kuliah (  
2     kode_mk VARCHAR(10) PRIMARY KEY,  
3     nama_mk VARCHAR(100) NOT NULL,  
4     sks INT NOT NULL,  
5     semester INT NOT NULL,  
6     nidn VARCHAR(10),  
7     FOREIGN KEY (nidn) REFERENCES dosen(nidn)  
8 );
```

- Creating a **KRS** table

```
1 CREATE TABLE krs (  
2     id_krs INT AUTO_INCREMENT PRIMARY KEY,  
3     nim VARCHAR(10),  
4     kode_mk VARCHAR(10),  
5     nilai VARCHAR(2),  
6     FOREIGN KEY (nim) REFERENCES mahasiswa(nim),  
7     FOREIGN KEY (kode_mk) REFERENCES mata_kuliah(kode_mk)  
8 );
```

- Creating a **Users** table

```
1 CREATE TABLE users (  
2     username VARCHAR(20) PRIMARY KEY,  
3     password VARCHAR(100),  
4     role ENUM('admin','mahasiswa'),  
5     nim VARCHAR(20),  
6     FOREIGN KEY (nim) REFERENCES mahasiswa(nim)  
7 );
```

MYSQL DATABASE CONFIGURATION

3. Once all the tables have been created, start filling in the data:

- Filling in Data to the **mahasiswa** table

```
1 INSERT INTO mahasiswa (nim, nama, jurusan, angkatan, no_hp) VALUES
2 ('2023001', 'Andi Saputra', 'Informatika', '2023', '081234567890'),
3 ('2023002', 'Budi Santoso', 'Informatika', '2023', '082134567891'),
4 ('2023003', 'Citra Lestari', 'Informatika', '2023', '083134567892');
```

- Filling in Data to the **dosen** table

```
1 INSERT INTO dosen (nidn, nama, prodi) VALUES
2 ('D001', 'Dr. Suryana', 'Informatika'),
3 ('D002', 'Dr. Ratna Dewi', 'Informatika');
```

- Filling in Data to the **mata_kuliah** table

First semester courses:

```
1 INSERT INTO mata_kuliah (kode_mk, nama_mk, sks, semester, nidn) VALUES
2 ('IF101', 'Pemrograman Dasar', 3, 1, 'D001'),
3 ('IF102', 'Matematika Diskrit', 3, 1, 'D002');
```

Second semester courses:

```
1 INSERT INTO mata_kuliah (kode_mk, nama_mk, sks, semester, nidn) VALUES
2 ('IF201', 'Struktur Data', 3, 2, 'D001'),
3 ('IF202', 'Basis Data', 3, 2, 'D002');
```

MYSQL DATABASE CONFIGURATION

- Filling in Data to the **KRS** table

Semester 1 credit value:

```
1 INSERT INTO krs (nim, kode_mk, nilai) VALUES
2 ('2023001', 'IF101', 'B'),
3 ('2023001', 'IF102', 'B+'),
4 ('2023002', 'IF101', 'A'),
5 ('2023002', 'IF102', 'B'),
6 ('2023003', 'IF101', 'A-'),
7 ('2023003', 'IF102', 'A');
```

Semester 2 credit value:

```
1 INSERT INTO krs (nim, kode_mk, nilai) VALUES
2 ('2023001', 'IF201', 'A'),
3 ('2023001', 'IF202', 'B+'),
4 ('2023002', 'IF201', 'B'),
5 ('2023002', 'IF202', 'B+'),
6 ('2023003', 'IF201', 'A'),
7 ('2023003', 'IF202', 'A-');
```

- Filling in Data to the **users** table

users for admin:

```
1 INSERT INTO users (username, password, role) VALUES
2 ('admin', 'admin123', 'admin');
3
```

Users for student:

```
1 INSERT INTO users (username, password, role, nim) VALUES
2 ('2023001', '2023001', 'mahasiswa', '2023001'),
3 ('2023002', '2023002', 'mahasiswa', '2023002'),
4 ('2023003', '2023003', 'mahasiswa', '2023003');
```

SOURCE CODE

System Process :

```
import mysql.connector
```

```
=====
```

Database Connection

```
=====
```

```
db = mysql.connector.connect(  
    host="localhost",  
    user="root",    # change according to MySQL user  
    password="",    # change according to MySQL password  
    database="akademik"  
)  
cursor = db.cursor()
```

```
=====
```

LOGIN

```
=====
```

```
def login():  
    print("\n=== SISTEM AKADEMIK ===")  
    print("1. Login Admin")  
    print("2. Login Mahasiswa")  
    print("0. Keluar")  
    pilihan = input("Pilih login sebagai (1/2/0): ")  
  
    if pilihan == "0":  
        print("Keluar...")  
        exit()  
  
    username = input("Username: ")  
    password = input("Password: ")  
    cursor.execute("SELECT * FROM users WHERE username=%s AND password=%s",  
(username, password))  
    user = cursor.fetchone()
```

SOURCE CODE

```
if not user:
    print("Login gagal! Cek username/password.")
    return None

role = user[3]
if pilihan == "1" and role == "admin":
    print(f"Login berhasil sebagai Admin ({username})")
    return ("admin", user)
elif pilihan == "2" and role == "mahasiswa":
    print(f"Login berhasil sebagai Mahasiswa ({username})")
    return ("mahasiswa", user)
else:
    print("Role tidak sesuai dengan pilihan login.")
    return None
```

=====

Manage Students

=====

```
def menu_mahasiswa_admin():
    while True:
        print("\n--- KELOLA MAHASISWA ---")
        print("1. Lihat Data Mahasiswa")
        print("2. Tambah Mahasiswa")
        print("3. Update Mahasiswa")
        print("4. Hapus Mahasiswa")
        print("0. Kembali")
        pilih = input("Pilih: ")

        if pilih == "1":
            cursor.execute("SELECT * FROM mahasiswa")
            for m in cursor.fetchall():
                print(f"NIM: {m[0]} | Nama: {m[1]} | Jurusan: {m[2]} | Angkatan: {m[3]}")
```

SOURCE CODE

```
elif pilih == "2":
    nim = input("Masukkan NIM: ")
    nama = input("Masukkan Nama: ")
    jurusan = input("Masukkan Jurusan: ")
    angkatan = int(input("Masukkan Angkatan: "))
    cursor.execute("INSERT INTO mahasiswa VALUES (%s,%s,%s,%s)", (nim, nama,
jurusan, angkatan))
    cursor.execute("INSERT INTO users (username,password,role,nim) VALUES
(%s,%s,'mahasiswa',%s)",
                    (nim, nim, nim))
    db.commit()
    print("Mahasiswa berhasil ditambahkan!")

elif pilih == "3":
    nim = input("Masukkan NIM Mahasiswa yang mau diupdate: ")
    nama = input("Nama baru: ")
    jurusan = input("Jurusan baru: ")
    angkatan = int(input("Angkatan baru: "))
    cursor.execute("UPDATE mahasiswa SET nama=%s,jurusan=%s,angkatan=%s
WHERE nim=%s",
                    (nama, jurusan, angkatan, nim))
    db.commit()
    print("Data mahasiswa berhasil diupdate!")

elif pilih == "4":
    nim = input("Masukkan NIM Mahasiswa yang mau dihapus: ")
    try:
        cursor.execute("DELETE FROM mahasiswa WHERE nim=%s", (nim,))
        cursor.execute("DELETE FROM users WHERE nim=%s", (nim,))
        db.commit()
        print("Mahasiswa berhasil dihapus!")
    except mysql.connector.Error as e:
        print(f"Gagal menghapus: {e}")

elif pilih == "0":
    break
```

SOURCE CODE

Manage Lecturers

```
=====
Manage Lecturers
=====

def menu_dosen_admin():
    while True:
        print("\n--- KELOLA DOSEN ---")
        print("1. Lihat Data Dosen")
        print("2. Tambah Dosen")
        print("3. Update Dosen")
        print("4. Hapus Dosen")
        print("0. Kembali")
        pilih = input("Pilih: ")

        if pilih == "1":
            cursor.execute("SELECT * FROM dosen")
            for d in cursor.fetchall():
                print(f"NIDN: {d[0]} | Nama: {d[1]} | Prodi: {d[2]}")

        elif pilih == "2":
            nidn = input("Masukkan NIDN: ")
            nama = input("Masukkan Nama: ")
            prodi = input("Masukkan Prodi: ")
            cursor.execute("INSERT INTO dosen VALUES (%s,%s,%s)", (nidn, nama, prodi))
            cursor.execute("INSERT INTO users (username,password,role,nidn) VALUES
(%s,%s,'dosen',%s)",
                (nidn, nidn, nidn))
            db.commit()
            print("Dosen berhasil ditambahkan!")

        elif pilih == "3":
            nidn = input("Masukkan NIDN yang mau diupdate: ")
            nama = input("Nama baru: ")
            prodi = input("Prodi baru: ")
            cursor.execute("UPDATE dosen SET nama=%s,prodi=%s WHERE nidn=%s", (nama,
prodi, nidn))
            db.commit()
            print("Data dosen berhasil diupdate!")
```


SOURCE CODE

```
        elif pilih == "4":
            nidn = input("Masukkan NIDN yang mau dihapus: ")
            try:
                cursor.execute("DELETE FROM dosen WHERE nidn=%s", (nidn,))
                cursor.execute("DELETE FROM users WHERE nidn=%s", (nidn,))
                db.commit()
                print("Dosen berhasil dihapus!")
            except mysql.connector.Error as e:
                print(f"Gagal menghapus: {e}")

        elif pilih == "0":
            break
        else:
            print("Pilihan tidak valid.")

=====
Manage Courses
=====

def menu_matkul_admin():
    while True:
        print("\n--- KELOLA MATA KULIAH ---")
        print("1. Lihat Mata Kuliah")
        print("2. Tambah Mata Kuliah")
        print("3. Update Mata Kuliah")
        print("4. Hapus Mata Kuliah")
        print("0. Kembali")
        pilih = input("Pilih: ")

        if pilih == "1":
            cursor.execute("SELECT * FROM matkul")
            for mk in cursor.fetchall():
                print(f"Kode: {mk[0]} | Nama: {mk[1]} | SKS: {mk[2]} | NIDN: {mk[3]}")
                {r['semester']} | Dosen: {r['nama_dosen']})
```

SOURCE CODE

```
elif pilih == "2":
    kode = input("Kode MK: ")
    nama = input("Nama MK: ")
    sks = int(input("SKS: "))
    nidn = input("NIDN Dosen: ")
    cursor.execute("INSERT INTO matkul VALUES (%s,%s,%s,%s)", (kode, nama, sks,
nidn))
    db.commit()
    print("Mata kuliah berhasil ditambahkan!")

elif pilih == "3":
    kode = input("Kode MK yang mau diupdate: ")
    nama = input("Nama baru: ")
    sks = int(input("SKS baru: "))
    nidn = input("NIDN baru: ")
    cursor.execute("UPDATE matkul SET nama_mk=%s,sks=%s,nidn=%s WHERE
kode_mk=%s",
                    (nama, sks, nidn, kode))
    db.commit()
    print("Mata kuliah berhasil diupdate!")

elif pilih == "4":
    kode = input("Kode MK yang mau dihapus: ")
    cursor.execute("DELETE FROM matkul WHERE kode_mk=%s", (kode,))
    db.commit()
    print("Mata kuliah berhasil dihapus!")

elif pilih == "0":
    break
else:
    print("Pilihan tidak valid.")
```

SOURCE CODE

Manage KRS

```
def menu_krs_admin():
    while True:
        print("\n--- KELOLA KRS ---")
        print("1. Lihat Data KRS")
        print("2. Tambah KRS")
        print("3. Update KRS")
        print("4. Hapus KRS")
        print("0. Kembali")
        pilih = input("Pilih: ")

        if pilih == "1":
            cursor.execute("SELECT * FROM krs")
            for k in cursor.fetchall():
                print(f"NIM: {k[1]} | Kode MK: {k[2]} | Semester: {k[3]} | Nilai: {k[4]}")

        elif pilih == "2":
            nim = input("Masukkan NIM: ")
            kode = input("Masukkan Kode Matkul: ")
            try:
                semester = int(input("Masukkan Semester (angka): "))
            except ValueError:
                print("Semester harus berupa angka!")
                continue
            nilai = input("Masukkan Nilai: ")
            cursor.execute("INSERT INTO krs (nim,kode_mk,semester,nilai) VALUES
(%s,%s,%s,%s)",
                (nim, kode, semester, nilai))
            db.commit()
            print("KRS berhasil ditambahkan!")
```

SOURCE CODE

```
elif pilih == "3":
    nim = input("Masukkan NIM: ")
    kode = input("Masukkan Kode Matkul: ")
    try:
        semester = int(input("Masukkan Semester (angka): "))
    except ValueError:
        print("Semester harus berupa angka!")
        continue
    nilai = input("Nilai baru: ")
    cursor.execute("UPDATE krs SET nilai=%s WHERE nim=%s AND kode_mk=%s
AND semester=%s",
                    (nilai, nim, kode, semester))
    db.commit()
    print("KRS berhasil diupdate!")

elif pilih == "4":
    nim = input("Masukkan NIM: ")
    kode = input("Masukkan Kode Matkul: ")
    try:
        semester = int(input("Masukkan Semester (angka): "))
    except ValueError:
        print("Semester harus berupa angka!")
        continue
    cursor.execute("DELETE FROM krs WHERE nim=%s AND kode_mk=%s AND
semester=%s",
                    (nim, kode, semester))
    db.commit()
    print("KRS berhasil dihapus!")

elif pilih == "0":
    break
else:
    print("Pilihan tidak valid.")
```

SOURCE CODE

Admin Menu

```
def menu_admin():
    while True:
        print("\n--- MENU ADMIN ---")
        print("1. Kelola Mahasiswa")
        print("2. Kelola Dosen")
        print("3. Kelola Mata Kuliah")
        print("4. Kelola KRS")
        print("0. Logout")
        pilih = input("Pilih: ")

        if pilih == "1":
            menu_mahasiswa_admin()
        elif pilih == "2":
            menu_dosen_admin()
        elif pilih == "3":
            menu_matkul_admin()
        elif pilih == "4":
            menu_krs_admin()
        elif pilih == "0":
            break
        else:
            print("Pilihan tidak valid.")
```

Student Menu

```
def menu_mahasiswa(user):
    nim = user[4]
    cursor.execute("SELECT * FROM mahasiswa WHERE nim=%s", (nim,))
    mhs = cursor.fetchone()
    print(f"\nSelamat datang, {mhs[1]} (NIM: {mhs[0]})")
```

SOURCE CODE

```
while True:
    print("\n--- MENU MAHASISWA ---")
    print("1. Lihat Profil")
    print("2. Lihat Nilai per Semester")
    print("3. Lihat Mata Kuliah")
    print("4. Lihat Data Dosen")
    print("0. Logout")
    pilih = input("Pilih: ")

    if pilih == "1":
        print(f"NIM: {mhs[0]} | Nama: {mhs[1]} | Jurusan: {mhs[2]} | Angkatan: {mhs[3]}")
    elif pilih == "2":
        try:
            semester = int(input("Masukkan Semester: "))
        except ValueError:
            print("Semester harus berupa angka!")
            continue
        cursor.execute("""
            SELECT matkul.nama_mk, matkul.sks, krs.nilai
            FROM krs
            JOIN matkul ON krs.kode_mk = matkul.kode_mk
            WHERE krs.nim=%s AND krs.semester=%s
            """, (nim, semester))
        hasil = cursor.fetchall()
        if not hasil:
            print("Belum ada nilai.")
        else:
            print(f"\n--- Nilai Semester {semester} ---")
            for row in hasil:
                print(f"{row[0]} ({row[1]} SKS) -> Nilai: {row[2]}")
    elif pilih == "3":
        cursor.execute("SELECT * FROM matkul")
        for mk in cursor.fetchall():
            print(f"Kode: {mk[0]} | Nama: {mk[1]} | SKS: {mk[2]} | NIDN: {mk[3]}")
```

SOURCE CODE

```
elif pilih == "4":
    cursor.execute("SELECT * FROM dosen")
    for dsn in cursor.fetchall():
        print(f"NIDN: {dsn[0]} | Nama: {dsn[1]} | Prodi: {dsn[2]}")
elif pilih == "0":
    break
else:
    print("Pilihan tidak valid.")
```

Main Program

```
while True:
    user_login = login()
    if user_login:
        role, user = user_login
        if role == "admin":
            menu_admin()
        elif role == "mahasiswa":
            menu_mahasiswa(user)
```

OUTPUT PROGRAM

Connect MySQL to Pycharm:

The crucial first step is to ensure the availability of the connector module. The command **pip install mysql-connector-python** is executed through the terminal. This command acts as a bridge that allows Python code to communicate with the MySQL server. Once executed, if the library is already installed, the program will confirm its status with the message **Requirement already satisfied: mysql-connector-python....** This confirmation ensures that the connector library is ready, allowing the use of the **import mysql.connector** command in the program code to establish a connection to the academic database.

```
Terminal Local x + v
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

(.venv) PS C:\Users\Muhammad Rezky Amri\PycharmProjects\PythonProject1> pip install mysql-connector-python
Requirement already satisfied: mysql-connector-python in c:\users\muhammad rezky amri\pycharmprojects\pythonproject1\.venv\lib\site-packages (9.4.0)

[notice] A new release of pip is available: 25.1.1 -> 25.2
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\Muhammad Rezky Amri\PycharmProjects\PythonProject1> 
```

Main Menu and Login:

- **Admin Login:**

If an administrator wants to log in, they will select option 1. After that, they will be asked to enter their username and password. If the credentials entered are correct, the program will display the message “Login berhasil sebagai Admin (admin)” and direct them straight to the admin menu.

```

=== SISTEM AKADEMIK ===
1. Login Admin
2. Login Mahasiswa
0. Keluar
Pilih login sebagai (1/2/0): 1
Username: admin
Password: admin123
Login berhasil sebagai Admin (admin)

```


OUTPUT PROGRAM

- **Student Login:**

If a student wants to log in, they will select option 2. They will be asked to enter their student ID number and password. If the student ID number entered matches the data in the database, the program will display a message, such as “Selamat datang, Andi Wijaya (NIM: 2023001)”.

```
=== SISTEM AKADEMIK ===
1. Login Admin
2. Login Mahasiswa
0. Keluar
Pilih login sebagai (1/2/0): 2
Username: 2023001
Password: 2023001

Selamat datang, Andi Wijaya (NIM: 2023001)
```

- **Failed Login:**

However, if there is an error in either the username or password, the program will deny access and display the message “Login gagal! Cek username/password.”

```
=== SISTEM AKADEMIK ===
1. Login Admin
2. Login Mahasiswa
0. Keluar
Pilih login sebagai (1/2/0): 1
Username: admin123
Password: admin123
Login gagal! Cek username/password.
```

OUTPUT PROGRAM

Admin Menu Functionality:

After successfully logging in, an administrator will see a menu with options for managing data.

1. Main Admin Menu

If the administrator selects one of the options from the main menu, the program will display a submenu that provides complete CRUD functionality.

```
--- MENU ADMIN ---
1. Kelola Mahasiswa
2. Kelola Dosen
3. Kelola Mata Kuliah
4. Kelola KRS
0. Logout
```

2. Manage Students

- Viewing Student Data

If the administrator selects option 1, View Student Data, the program will run a SELECT query to retrieve and detail all student data in the output, showing the student ID number, name, department, and class year of each student.

```
--- KELOLA MAHASISWA ---
1. Lihat Data Mahasiswa
2. Tambah Mahasiswa
3. Update Mahasiswa
4. Hapus Mahasiswa
0. Kembali
Pilih: 1
NIM: 2023001 | Nama: Andi Wijaya | Jurusan: Informatika | Angkatan: 2023
NIM: 2023002 | Nama: Budi Santoso | Jurusan: Informatika | Angkatan: 2023
NIM: 2023003 | Nama: Citra Lestari | Jurusan: Informatika | Angkatan: 2023
NIM: 2023004 | Nama: asep | Jurusan: informatika | Angkatan: 2024
NIM: 2023005 | Nama: ucok | Jurusan: informatika | Angkatan: 2024
NIM: 2023006 | Nama: cecep | Jurusan: informatika | Angkatan: 2023
NIM: 2023007 | Nama: paris | Jurusan: informatika | Angkatan: 2024
```

OUTPUT PROGRAM

- Creating Student Data

If the administrator selects option “2. Tambah Mahasiswa”, the program will guide the administrator to enter the new student's details, and if the INSERT query is successfully executed, the program will confirm: “Mahasiswa berhasil ditambahkan!”.

```
--- KELOLA MAHASISWA ---
1. Lihat Data Mahasiswa
2. Tambah Mahasiswa
3. Update Mahasiswa
4. Hapus Mahasiswa
0. Kembali
Pilih: 2
Masukkan NIM: 2023008
Masukkan Nama: mesi
Masukkan Jurusan: informatika
Masukkan Angkatan: 2025
Mahasiswa berhasil ditambahkan!
```

- Update Student Data

If the administrator selects option “3. Update Mahasiswa”, the program will request the student ID number and the new details to be changed; if the UPDATE query is successfully executed, the program will confirm with the message: “Data mahasiswa berhasil diupdate!”.

```
--- KELOLA MAHASISWA ---
1. Lihat Data Mahasiswa
2. Tambah Mahasiswa
3. Update Mahasiswa
4. Hapus Mahasiswa
0. Kembali
Pilih: 3
Masukkan NIM Mahasiswa yang mau diupdate: 2023008
Nama baru: yamal
Jurusan baru: informatika
Angkatan baru: 2024
Data mahasiswa berhasil diupdate!
```

OUTPUT PROGRAM

- Deleting Student Data

If the administrator selects option “4. Hapus Mahasiswa” and enters the student ID number, the program will delete the data from the database using a DELETE query and confirm with the message: “Mahasiswa dan data terkait berhasil dihapus!”

```
--- KELOLA MAHASISWA ---
1. Lihat Data Mahasiswa
2. Tambah Mahasiswa
3. Update Mahasiswa
4. Hapus Mahasiswa
0. Kembali
Pilih: 4
Masukkan NIM Mahasiswa yang mau dihapus: 2023008
Mahasiswa dan data terkait berhasil dihapus!
```

3. Manage Lecturers

- Viewing Faculty Data

If the administrator selects option “1. Lihat Data Dosen”, the program will run a SELECT query to retrieve and detail all lecturer data in the output, showing the NIDN, name, and study program of each lecturer.

```
--- KELOLA DOSEN ---
1. Lihat Data Dosen
2. Tambah Dosen
3. Update Dosen
4. Hapus Dosen
0. Kembali
Pilih: 1
NIDN: D001 | Nama: Dr. Suryana | Prodi: Informatika
NIDN: D002 | Nama: Dr. Ratna Dewi | Prodi: Informatika
NIDN: D003 | Nama: udin | Prodi: Informatika
NIDN: D004 | Nama: nay | Prodi: informatika
```

OUTPUT PROGRAM

- Creating Faculty Data

If the administrator selects option “2. Tambah Dosen”, the program will guide the administrator to enter the NIDN, Name, and Study Program of the new lecturer. If the INSERT query is successfully executed, the program will confirm: “Dosen berhasil ditambahkan!”.

```
--- KELOLA DOSEN ---
1. Lihat Data Dosen
2. Tambah Dosen
3. Update Dosen
4. Hapus Dosen
0. Kembali
Pilih: 2
Masukkan NIDN: 0005
Masukkan Nama: mile
Masukkan Prodi: informatika
Dosen berhasil ditambahkan!
```

- Update Faculty Data

If the administrator selects option “3. Update Dosen”, the program will request the NIDN and new details. If the UPDATE query is successfully executed, the program will confirm with the message: “Data dosen berhasil diupdate!”.

```
--- KELOLA DOSEN ---
1. Lihat Data Dosen
2. Tambah Dosen
3. Update Dosen
4. Hapus Dosen
0. Kembali
Pilih: 3
Masukkan NIDN yang mau diupdate: 0004
Nama baru: naysilla
Prodi baru: informatika
Data dosen berhasil diupdate!
```

OUTPUT PROGRAM

- Deleting Faculty Data

If the administrator selects option “4. Hapus Dosen” and enters the NIDN, the program will delete the data from the database using the DELETE query and confirm with the message: “Dosen berhasil dihapus!”.

```
--- KELOLA DOSEN ---
1. Lihat Data Dosen
2. Tambah Dosen
3. Update Dosen
4. Hapus Dosen
0. Kembali
Pilih: 4
Masukkan NIDN yang mau dihapus: 0004
Dosen berhasil dihapus!
```

4. Manage Courses

- Viewing Courses Data

If the administrator selects option “1. Lihat Mata Kuliah”, the program will run a SELECT query to retrieve and detail all course data in the database.

```
--- KELOLA MATA KULIAH ---
1. Lihat Mata Kuliah
2. Tambah Mata Kuliah
3. Update Mata Kuliah
4. Hapus Mata Kuliah
0. Kembali
Pilih: 1
Kode: IF101 | Nama: Pemrograman Dasar | SKS: 3 | NIDN: D001
Kode: IF102 | Nama: Matematika Diskrit | SKS: 3 | NIDN: D001
Kode: IF201 | Nama: Python | SKS: 3 | NIDN: D001
Kode: IF301 | Nama: Algoritma | SKS: 3 | NIDN: D003
```

OUTPUT PROGRAM

- Creating Courses Data

If the administrator selects option “2. Tambah Matkul” and successfully enters the details, the program will run the INSERT query and confirm the successful addition of data.

```
--- KELOLA MATA KULIAH ---
1. Lihat Data Matkul
2. Tambah Matkul
3. Update Matkul
4. Hapus Matkul
0. Kembali
Pilih: 2
Masukkan Kode Matkul: IF301
Masukkan Nama Matkul: Algoritma
Masukkan SKS: 3
Masukkan NIDN Dosen: 0003
Matkul berhasil ditambahkan!
```

- Update Courses Data

If the administrator selects option “3. Update Matkul”, the program will request the Course Code and the new details to be changed; if the UPDATE query is successfully executed, the program will confirm with the message: “Mata kuliah berhasil diupdate!”.

```
--- KELOLA MATA KULIAH ---
1. Lihat Mata Kuliah
2. Tambah Mata Kuliah
3. Update Mata Kuliah
4. Hapus Mata Kuliah
0. Kembali
Pilih: 3
Kode MK yang mau diupdate: IF301
Nama baru: Algoritma
SKS baru: 3
NIDN baru: 0003
Mata kuliah berhasil diupdate!
```

OUTPUT PROGRAM

- Deleting Courses Data

If the administrator selects option “4. Hapus Mata Kuliah” and enters the course code to be deleted, the program will delete the data from the database using a DELETE query and confirm with the message: “Mata kuliah berhasil dihapus!”.

```
--- KELOLA MATA KULIAH ---
1. Lihat Mata Kuliah
2. Tambah Mata Kuliah
3. Update Mata Kuliah
4. Hapus Mata Kuliah
0. Kembali
Pilih: 4
Kode MK yang mau dihapus: IF301
Mata kuliah berhasil dihapus!
```

5. Manage Course Selection Sheet(KRS)

- Viewing KRS Data

If the administrator selects option “1. Lihat Data KRS”, it will display all NIM data, Course Codes, Semesters, and Grades from the database to retrieve and detail all course data in the database.

```
--- KELOLA KRS ---
1. Lihat Data KRS
2. Tambah KRS
3. Update KRS
4. Hapus KRS
0. Kembali
Pilih: 1
NIM: 2023001 | Kode MK: IF101 | Semester: 1 | Nilai: B
NIM: 2023001 | Kode MK: IF102 | Semester: 1 | Nilai: B+
NIM: 2023001 | Kode MK: IF201 | Semester: 2 | Nilai: A
NIM: 2023002 | Kode MK: IF101 | Semester: 1 | Nilai: C+
```


OUTPUT PROGRAM

- Creating and Update KRS Data

If the administrator selects option “2. Tambah KRS” and “3. Update KRS”, the program will request the student ID number, course code, semester, and grade, and if the INSERT query is successfully executed, the program will confirm the addition of the KRS.

```
--- KELOLA KRS ---
1. Lihat Data KRS
2. Tambah KRS
3. Update KRS
4. Hapus KRS
0. Kembali
Pilih: 2
Masukkan NIM: 2023002
Masukkan Kode Matkul: IF102
Masukkan Semester (angka): 1
Masukkan Nilai: A-
KRS berhasil ditambahkan!
```

- Deleting KRS Data

If the administrator selects option “4. Hapus KRS”, the program will request the student ID number, course code, and semester as the keys for the data to be deleted. If the DELETE query is successfully executed, the program will confirm the deletion of the KRS data.

```
--- KELOLA KRS ---
1. Lihat Data KRS
2. Tambah KRS
3. Update KRS
4. Hapus KRS
0. Kembali
Pilih: 4
Masukkan NIM: 2023002
Masukkan Kode Matkul: IF101
Masukkan Semester (angka): 1
KRS berhasil dihapus!
```

OUTPUT PROGRAM

Student Menu Functionality:

For students who have successfully logged in, the menu displayed will focus on personal information.

1. View Profil

If students select option “1. Lihat Profil”, the program will retrieve and display their personal profile data (student ID number, name, major, class year).

```
--- MENU MAHASISWA ---
1. Lihat Profil
2. Lihat Nilai per Semester
3. Lihat Mata Kuliah
4. Lihat Data Dosen
0. Logout
Pilih: 1
NIM: 2023001 | Nama: Andi Wijaya | Jurusan: Informatika | Angkatan: 2023
```

2. Viewing Grades per Semester

If students choose option “2. Lihat Nilai per Semester”, the program will ask for semester input, and if Semester 1 or Semester 2 is entered, the program will display a list of courses and grades that students have obtained in that semester.

```
--- MENU MAHASISWA ---
1. Lihat Profil
2. Lihat Nilai per Semester
3. Lihat Mata Kuliah
4. Lihat Data Dosen
0. Logout
Pilih: 2
Masukkan Semester: 1

--- Nilai Semester 1 ---
Pemrograman Dasar (3 SKS) -> Nilai: B
Matematika Diskrit (3 SKS) -> Nilai: B+

--- MENU MAHASISWA ---
1. Lihat Profil
2. Lihat Nilai per Semester
3. Lihat Mata Kuliah
4. Lihat Data Dosen
0. Logout
Pilih: 2
Masukkan Semester: 2

--- Nilai Semester 2 ---
Struktur Data (3 SKS) -> Nilai: A
```

OUTPUT PROGRAM

3. Viewing Courses

If students select option “3. Lihat Mata Kuliah”, the program will display a list of available courses (Code, Name, Credits, NIDN).

```
--- MENU MAHASISWA ---
1. Lihat Profil
2. Lihat Nilai per Semester
3. Lihat Mata Kuliah
4. Lihat Data Dosen
0. Logout
Pilih: 3
Kode: IF101 | Nama: Pemrograman Dasar | SKS: 3 | NIDN: D001
Kode: IF102 | Nama: Matematika Diskrit | SKS: 3 | NIDN: D001
Kode: IF201 | Nama: Struktur Data | SKS: 3 | NIDN: D002
Kode: IF202 | Nama: Basis Data | SKS: 3 | NIDN: D002
```

4. View Faculty Data

If they select option “4. Lihat Data Dosen”, the program will display a list of lecturer data.

```
--- MENU MAHASISWA ---
1. Lihat Profil
2. Lihat Nilai per Semester
3. Lihat Mata Kuliah
4. Lihat Data Dosen
0. Logout
Pilih: 4
NIDN: D001 | Nama: Dr. Suryana | Prodi: Informatika
NIDN: D002 | Nama: Dr. Ratna Dewi | Prodi: Informatika
NIDN: D003 | Nama: udin | Prodi: Informatika
NIDN: D005 | Nama: mile | Prodi: informatika
```

PROJECT DEVELOPER

Develop by : Muhammad Rezky Amri & Naysilla Khasanah
Hardware : Asus Vivobook & Lenovo loq 15
Operating System : Microsoft Window 11
Software : 1. XAMPP/PhpMyAdmin
2. Microsoft word
3. Microsoft PowerPoint
4. Github: <https://github.com/muhammadrezkyamri/Database-matters-regarding-lectures-become-easier-using-this-program>